

Q5 Smart Campus Parking & Access Management System.

Algorithms :-

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1. Start
2. Input N. While $N < 0$, ask again.
3. Set Used A, Used B, Used C, processed, Accepted, Rejected, Cars, Bikes and Vans to 0.
4. While processed $< N$:-
 1. Input Type. While it is neither C, B, nor V, ask again.
 2. Input Category. While it is neither F, S, nor G, ask again.
 3. Input Permit. While it is neither Y nor N, ask again.
 4. Input Emergency. While it is neither Y nor N, ask again.
 5. If Permit = "N" AND Emergency = "N":
 - Display "Rejected : Invalid Permit".
 - Increased Rejected by 1.

6. Otherwise:
1. If Emergency status is Y:
• Select A for faculty, B for students,
or C for guests.

2. Otherwise:
• Select A for faculty.
• For students, select C for vans and
B for cars or bikes.
• Select C for guests.

3. If Zone: "C" AND Type = "V", set spaces $\leftarrow 2$;
Otherwise, set spaces $\leftarrow 1$.

4. Calculate free spaces in the selected zones:

- A: Free $\leftarrow 20 - \text{Used A}$
- B: Free $\leftarrow 40 - \text{Used B}$
- C: Free $\leftarrow 15 - \text{Used C}$

5. If Free $\geq$ Spaces

- Add spaces to the selected zone's
occupied spaces.
- Increase Accepted by 1
- Increase cars for C, Bikes for B, or
Vans for V.
- Display the assigned zone and Free-spaces.

6. Otherwise,

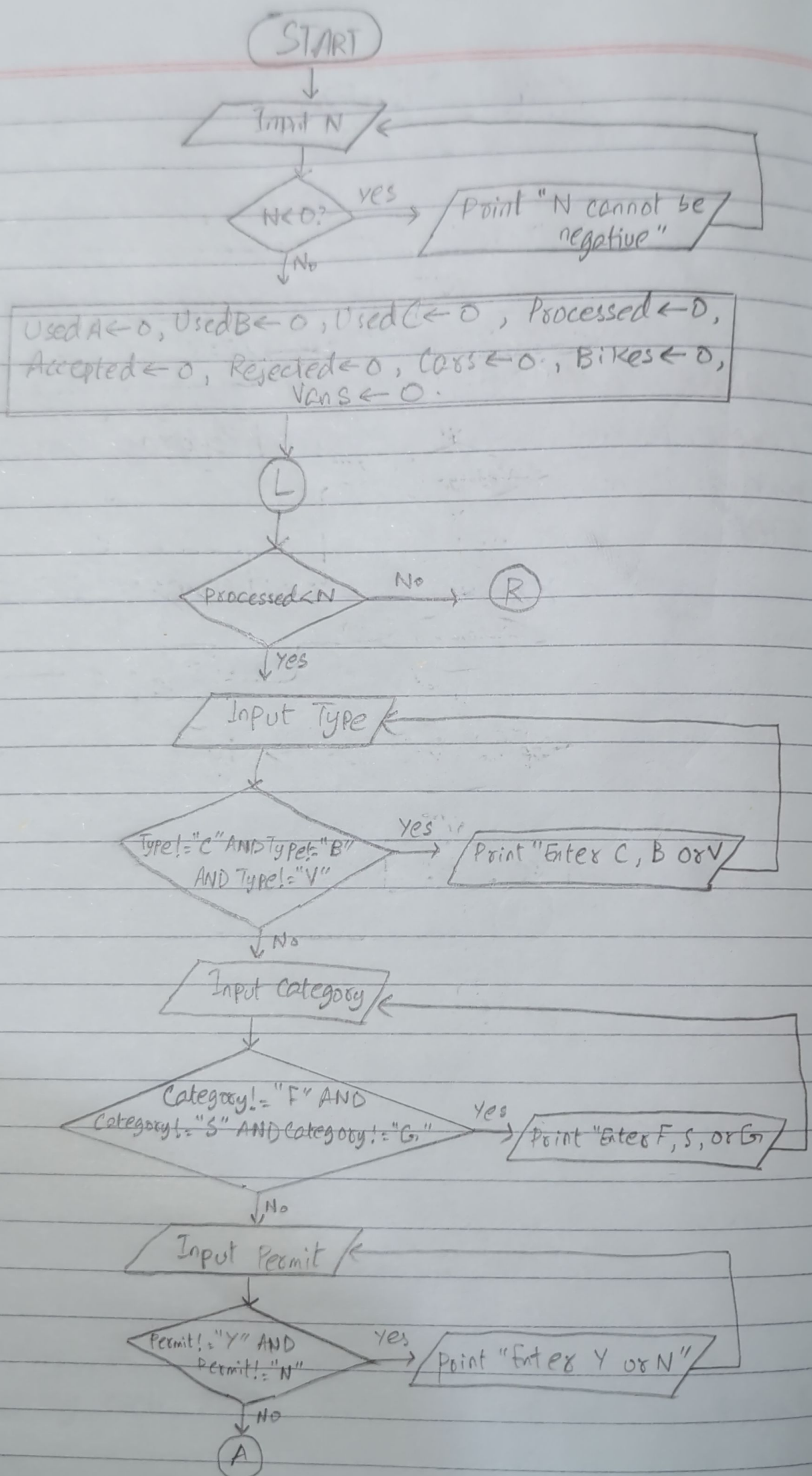
- Display "Rejected: not enough space."
- Increase Rejected by 1.

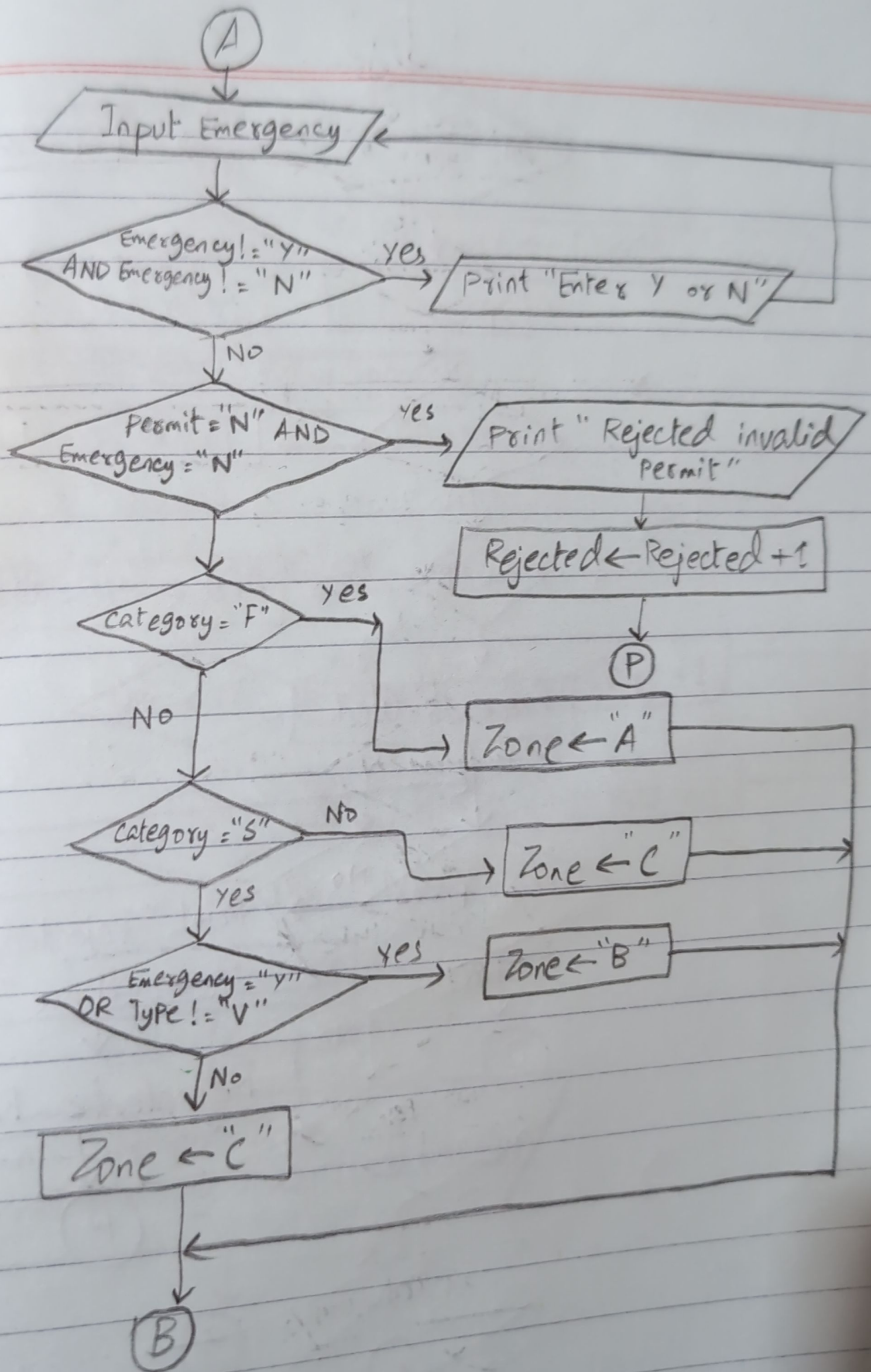
7. Increase Processed by 1.

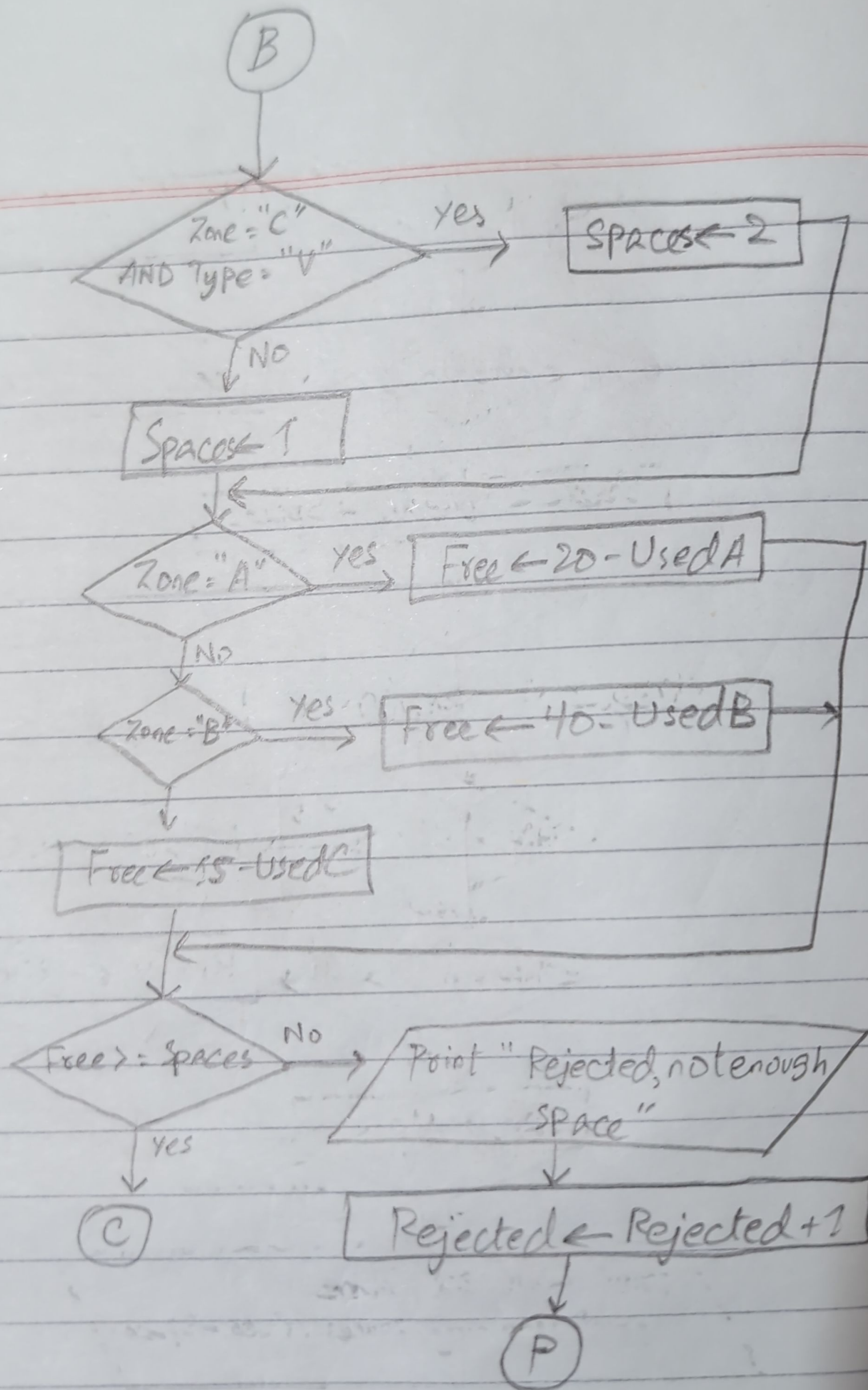
5. Display all vehicle counters.

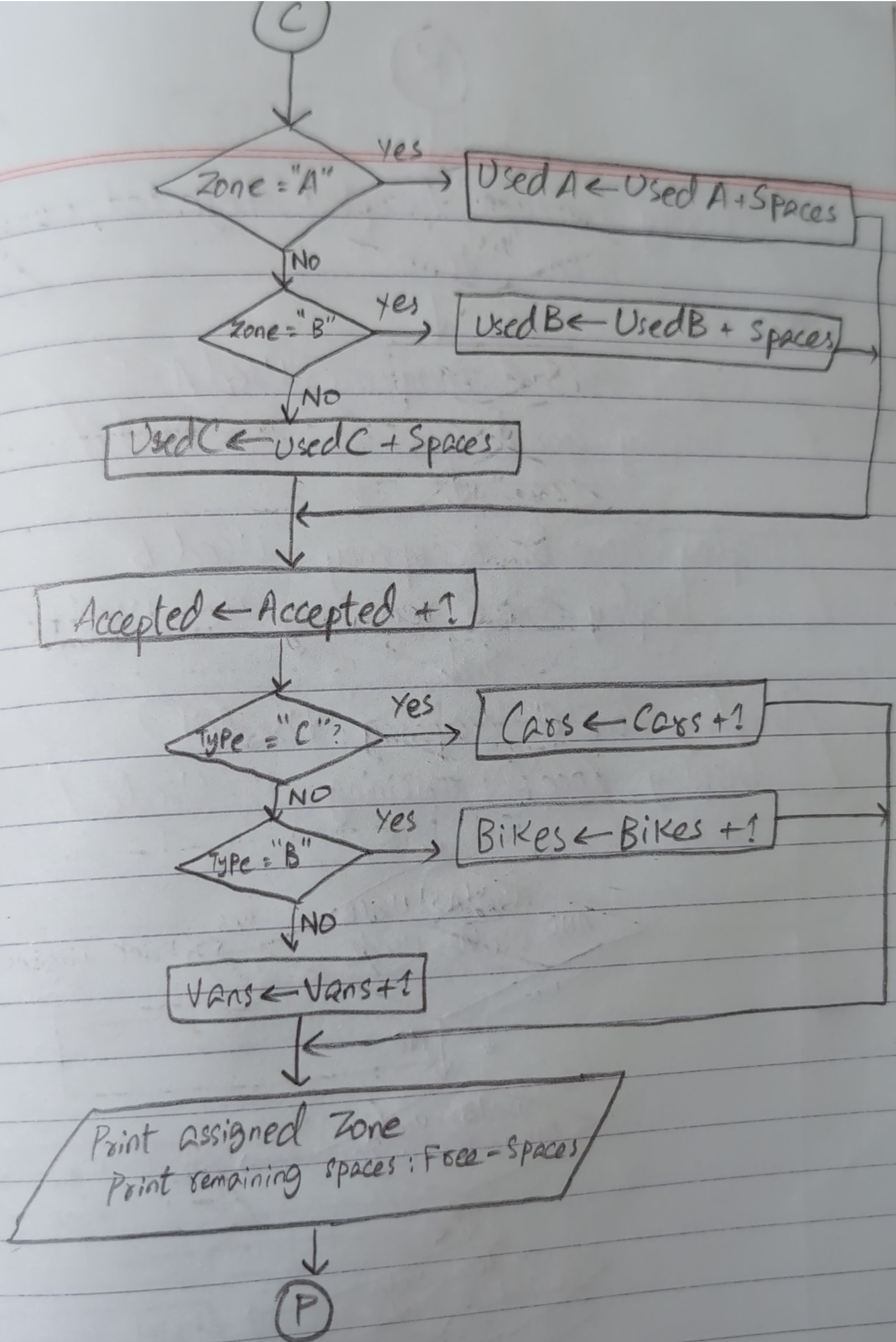
6. Display each zone's occupancy and remaining capacity.

7. If $UsedA \geq UsedB$ AND $UsedA \geq UsedC$, display A as a highest-occupancy zone.
8. If $UsedB \geq UsedA$ AND $UsedB \geq UsedC$, display B as a highest occupancy zone.
9. If $UsedC \geq UsedA$ AND $UsedC \geq UsedB$, display C as a highest occupancy zone.
10. If $UsedA = 20$ AND $UsedB = 40$ AND $UsedC = 15$, display "Campus parking is full"; otherwise display "Campus parking is not full".
11. STOP.





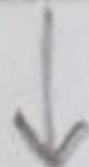




P



Processed $\leftarrow$ Processed + 1



L

(R)

Display Processed, Accepted, Rejected, Cars
Bikes, Vans

Display Zone A occupancy: Used A
Display Zone A remaining: 20 - Used A

Display Zone B occupancy: Used B
Display Zone B remaining: 40 - Used B

Display Zone C occupancy: Used C
Display Zone C remaining: 15 - Used C

Used A >= Used B
AND Used A >= Used C

Yes

Print "Highest occupancy:
Zone A"

NO

Used B >= Used A
AND Used B >= Used C

Yes

Print "Highest occupancy:
Zone B"

NO

Used C >= Used A
AND Used C >= Used B

Yes

Print "Highest occupancy:
Zone C"

NO

Used A = 20 AND
Used B = 40 AND
Used C = 15

Yes

Print "Campus parking is
full"

NO

Print "Campus parking
is not full"

STOP

Pseudo-code

Begin

Input N

While ($N < 0$), then
 point "Number of vehicles cannot be negative".
 Input N

END while.

Used A $\leftarrow 0$

Used B $\leftarrow 0$

Used C $\leftarrow 0$

Processed $\leftarrow 0$

Accepted $\leftarrow 0$

Rejected $\leftarrow 0$

Cars $\leftarrow 0$

Bikes $\leftarrow 0$

Vans $\leftarrow 0$

While (processed $< N$), do

 Input Type

 While (Type $\neq$ "C" AND Type $\neq$ "B" AND Type $\neq$ "V"), then

 Display "Enter C, B or V"

 Input Type

 END while

 Input Category

 While (Category $\neq$ "F" AND Category $\neq$ "S" ~~AND~~ AND
 Category $\neq$ "G"), do

 point "Enter F, S, or G"

 Input Category

 END while

```
Input Permit
While (Permit != "Y" AND Permit != "N"), do
    print "Enter Y or N"
    Input Permit
END While
```

```
Input Emergency
While (Emergency != "Y" AND Emergency != "N"), do
    print "Enter Y or N"
    Input Emergency
END While
```

```
IF (Permit = "N" AND Emergency = "N"), then
    print "Rejected: Invalid Permit"
    Rejected ← Rejected + 1
```

ELSE

```
IF (Emergency = "Y"), then
    IF (Category = "F"), then
        Zone ← "A"
    ELSE IF (Category = "S"), then
        Zone ← "B"
    ELSE
        Zone ← "C"
```

END IF

ELSE

```
IF (category = "F"), then
    Zone ← "A"
ELSE IF (Category = "S"), then
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IF (Type = "V"), then
Zone ← "C"

ELSE

Zone ← "B"

END IF

ELSE

Zone ← "C"

END IF

END IF

IF (Zone = "C" AND Type = "V"), then

Spaces ← 2

ELSE

Spaces ← 1

END IF

IF (Zone = "A"), then

Free ← 20 - Used A

ELSE IF (Zone = "B"), then

Free ← 40 - Used B

ELSE

Free ← 15 - Used C

END IF

IF (Free ≥ Spaces), then

IF (Zone = "A"), then

UsedA $\leftarrow$ UsedA + Spaces

ELSE IF (Zone = "B"), then

UsedB $\leftarrow$ UsedB + Spaces

ELSE

UsedC $\leftarrow$ UsedC + Spaces

END IF

Accepted $\leftarrow$ Accepted + 1

IF (Type = "C"), then

Cars $\leftarrow$ Cars + 1

ELSE IF (Type = "B"), then

Bikes $\leftarrow$ Bikes + 1

ELSE

Vans $\leftarrow$ Vans + 1

END IF

Display "Assigned Zone: ", Zone

Display "Remaining Spaces: ", Free - Spaces

ELSE

Display "Rejected: not enough Spaces"

Rejected $\leftarrow$ Rejected + 1

END IF

END IF

Processed $\leftarrow$ Processed + 1

ND. While

Display "Total Processed: ", Processed
Display "Total Accepted: ", Accepted
Display "Total Rejected: ", Rejected
Display "Cars Parked: ", Cars
Display "Bikes Parked: ", Bikes
Display "Vans Parked: ", Vans
Display "Zone A occupied: ", Used A
Display "Zone A remaining: ", 20 - Used A
Display "Zone B occupied: ", Used B
Display "Zone B remaining: ", 40 - Used B
Display "Zone C occupied: ", Used C
Display "Zone C remaining: ", 15 - Used C

IF (Used A $\geq$ Used B AND ~~Used A~~ Used A $\geq$ Used C), then
 print "Highest occupancy: Zone A"
END IF

IF (Used B $\geq$ Used A AND Used B $\geq$ Used C), then
 print "Highest occupancy: Zone B"
END IF

IF (Used C $\geq$ Used A AND Used C $\geq$ Used B), then
 print "Highest occupancy: Zone C"
END IF

IF (Used A = 20 AND Used B = 40 AND Used C = 15), then
 print "Campus Parking is Full"
~~END IF~~

ELSE

print "Campus Parking is not full"

END IF

STOP .
